Introduction:

This database models the operations of a sport management system. Teams, vendors, and attendees come together at various venues for scheduled events. Each entity in this model will possess unique identifying markers and engage in relationships with each other by either direct or indirect means.

Events serve as a central hub connecting all entities. As in the real world, events must have time, location, description, and attendees. The event itself will hold the time and description elements and through relationships we will derive the rest. Events will be held at venues that hold the information on location. Events will also allow attendees to register for attendance at the event, giving the event itself access to a list of attendees. There is no event without entertainment and vendors. Entertainment takes the form of a competitive match between two teams, that is modeled through a participation relationship. Vendors are assigned to different locations in a venue which is attached to the event when the venue is assigned.

Venues will have names and location information that allow users to locate venues and view information about them like capacity. Each venue will have locker rooms and vendor booth locations. Teams will be assigned a locker room where they can prepare for the match ahead. Vendors will be assigned locations that will allow them to set up and serve attendees. These venues will also have a capacity that tells the event how many attendees can attend any given event.

Attendees can register for events and information will be stored about the attendee to ensure that each attendee has proper access to their tickets and can receive updates on changes to the event like cancellation or rescheduling.

Teams participate in events and are assigned to locker rooms at the venue. Team entities contain the number of players that will participate, a team name, and manager contact information. This allows users to contact and coordinate assignments to the team.

Vendors will have a name, a type, and contact information. This allows the users to assign the vendors to a booth location in the venue and allows for communication with vendors in case of changes to the event like relocations, cancellation, or rescheduling.

Requirements analysis:

Our data models a complete sports event management system. It includes information about teams, vendors, attendees, venues, and the physical spaces within each venue,

such as locker rooms and vendor booths. Each event serves as the central organizing unit that connects all these entities. Every participating individual or group is represented with an ID, name, and contact information, while each location is identified by a location ID and name. Relationship tables capture the many-to-many links between events and their participants, ensuring that each event maintains an accurate record of who is involved and where they are assigned.

The system enforces constraints to prevent conflicts and maintain data integrity. A locker room may only be assigned to one team per event, and each vendor booth may only be assigned to a single vendor. While teams and vendors may participate in multiple events or appear at different venues, each event must occur at one venue, and each venue can host only one instance of that event. These rules ensure that assignments remain valid and non-overlapping.

The system supports operations for creating and managing events. When an event is created, users can assign teams to locker rooms, assign vendors to booths, and register attendees. Users can also view which teams are participating in an event, what events are scheduled at each venue, and how many attendees are registered for any given event. These users could be database admin, event managers, sports association, or even attendees.

ER diagram and constraints:

Overview:

The ER design for the sports event management system includes the entities of Team, Vendor, Event, Venue, Attendee, LockerRoom and VendorBooth. Even servers as the central entity connecting participants, registration and assignments.

Entity Sets:

- **Team**(team_id, team_name, manager_name, manager_contact, number_of_players)
- **Vendor(**vendor_id, vendor_name, contact_info, type)
- **Attendee**(attendee_id, attendee_name, email, phone_number)
- **Venue**(venue_id, venue_name, location_name, address, capacity)
- **LockerRoom**(locker_room_id, room_number)
- **VendorBooth**(booth_id, booth_number)
- **Event**(event_id, event_name, event_date, start_time, end_time, description)

Relationships:

- **HeldAt**(Event, Venue): represents where an event takes place
- **ContainsLockerRoom**(Venue, LockerRoom): represents the association between a venue and its locker rooms
- **ContainsBooth(Venue, VendorBooth)**: Represents the association between a venue and its vendor booths.
- **ParticipatesIn(Team, Event)**: Represents team participation in events.
- **SuppliesAt(Vendor, Event)**: Represents vendor participation in events.
- **RegistersFor(Attendee, Event)**: Represents attendee registration for events.
- **TeamAssignment(Team, Event, LockerRoom)**: Represents locker room assignments for teams within events.
- **VendorAssignment(Vendor, Event, VendorBooth)**: Represents booth assignments for vendors within events.

Formal Constraints:

Cardinality and participation constraints

1. Each Event must be held at exactly one Venue.
2. A Venue may host zero or more Events.
3. Each LockerRoom belongs to exactly one Venue.
4. A Venue may contain zero or more LockerRooms.
5. Each VendorBooth belongs to exactly one Venue.
6. A Venue may contain zero or more VendorBooths.
7. A Team may participate in zero or more Events, and an Event may include zero or more Teams.
8. A Vendor may participate in zero or more Events, and an Event may include zero or more Vendors.
9. An Attendee may register for zero or more Events, and an Event may have zero or more Attendees.

Assignment constraints

10. For a given Event, a LockerRoom may be assigned to at most one Team.
11. For a given Event, a Team may be assigned to at most one LockerRoom.
12. For a given Event, a VendorBooth may be assigned to at most one Vendor.
13. For a given Event, a Vendor may be assigned to at most one VendorBooth.

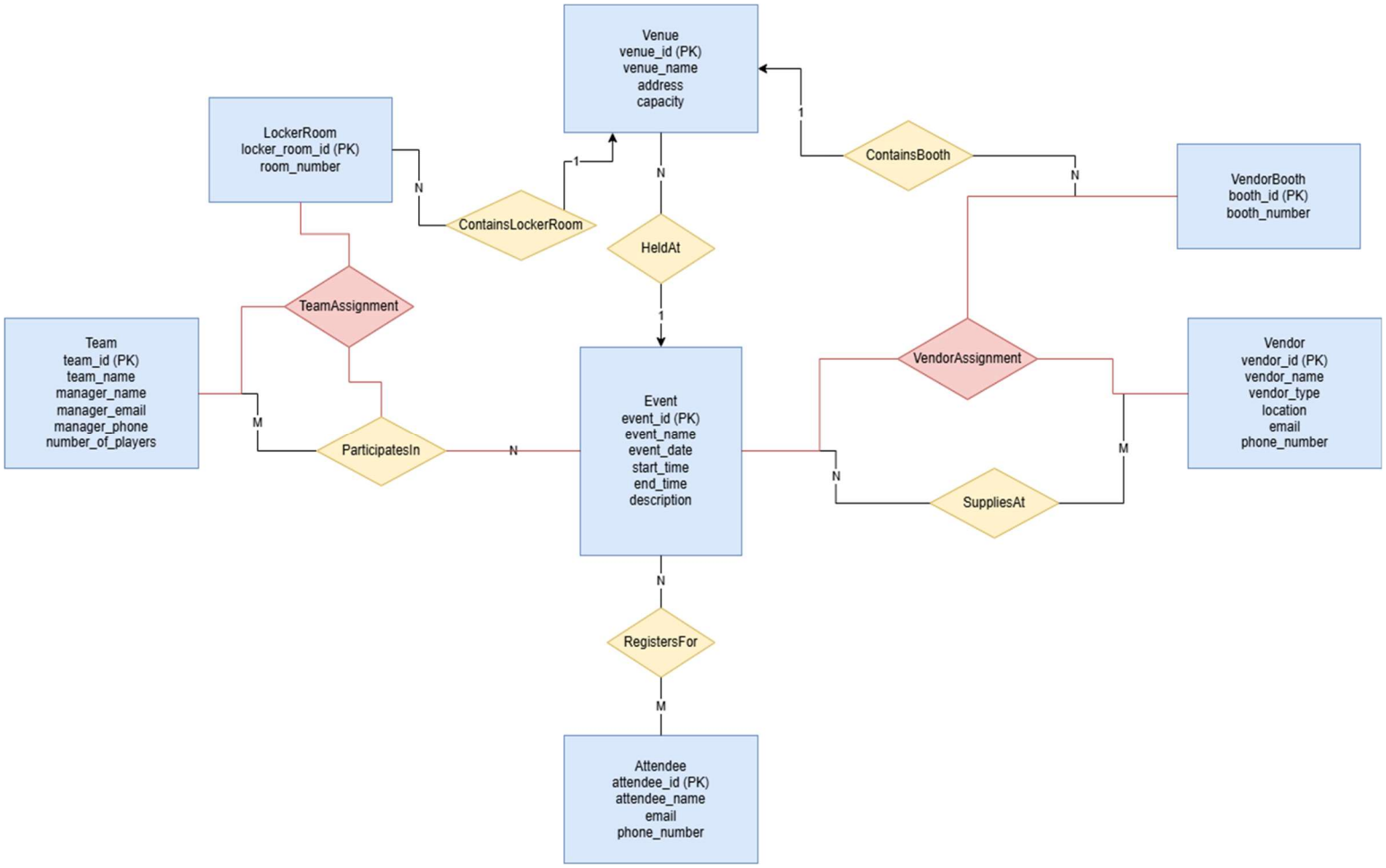
Venue-specific uniqueness constraints

14. Within a given Venue, each room_number must be unique.
15. Within a given Venue, each booth_number must be unique.

Semantic consistency constraints

16. A LockerRoom assigned to a Team for an Event must belong to the same Venue at which the Event is held.
17. A VendorBooth assigned to a Vendor for an Event must belong to the same Venue at which the Event is held.

ER Diagram:



The ER diagram uses color only for visual clarity. Blue rectangles represent entity sets, yellow diamonds represent regular binary relationship sets, and red diamonds represent ternary assignment relationship sets. The color differences are only used to make the structure easier to read and do not imply additional semantic meaning beyond the relationship type.

Logical Design:

The ER design was converted into relational schemas by applying the standard ER-to-relational mapping rules. Each strong entity set was mapped to its own relation, with its primary key becoming the primary key of the corresponding relation. One-to-many relationships were implemented by placing the primary key of the "one" side as a foreign key in the relation on the "many" side. For example, since each event is held at one venue, `venue_id` appears as a foreign key in `Event`. Similarly, because each locker room and vendor booth belongs to one venue, `venue_id` is included as a foreign key in `LockerRoom` and `VendorBooth`.

Many-to-many relationships were converted into separate relations whose primary keys are the combination of the participating entity keys. This applies to `ParticipatesIn`, `SuppliesAt`, and `RegistersFor`, which represent the associations between teams and events, vendors and events, and attendees and events, respectively. The ternary relationships of `TeamAssignment` and `VendorAssignment` were also converted into separate relations. These relations include the keys of the participating entities and the assigned locker room or booth, allowing the database to represent event-specific team and vendor assignments while preserving the constraints defined in the ER model.

Relational Schemas:

Team(`team_id`, team_name, manager_name, manager_email, manager_phone, number_of_players)
PK: `team_id`

Vendor(`vendor_id`, vendor_name, vendor_type, location, email, phone_number)
PK: `vendor_id`

Attendee(`attendee_id`, attendee_name, email, phone_number)
PK: `attendee_id`

Venue(`venue_id`, venue_name, address, capacity)
PK: `venue_id`

LockerRoom(locker_room_id, room_number, venue_id)
PK: locker_room_id
FK: venue_id → Venue(venue_id)
UNIQUE: (venue_id, room_number)

VendorBooth(booth_id, booth_number, venue_id)
PK: booth_id
FK: venue_id → Venue(venue_id)
UNIQUE: (venue_id, booth_number)

Event(event_id, event_name, event_date, start_time, end_time, description, venue_id)
PK: event_id
FK: venue_id → Venue(venue_id)

ParticipatesIn(team_id, event_id)
PK: (team_id, event_id)
FK: team_id → Team(team_id)
FK: event_id → Event(event_id)

SuppliesAt(vendor_id, event_id)
PK: (vendor_id, event_id)
FK: vendor_id → Vendor(vendor_id)
FK: event_id → Event(event_id)

RegistersFor(attendee_id, event_id)
PK: (attendee_id, event_id)
FK: attendee_id → Attendee(attendee_id)
FK: event_id → Event(event_id)

TeamAssignment(team_id, event_id, locker_room_id)
PK: (team_id, event_id)
FK: team_id → Team(team_id)
FK: event_id → Event(event_id)
FK: locker_room_id → LockerRoom(locker_room_id)
UNIQUE: (event_id, locker_room_id)

VendorAssignment(vendor_id, event_id, booth_id)
PK: (vendor_id, event_id)
FK: vendor_id → Vendor(vendor_id)
FK: event_id → Event(event_id)

FK: `booth_id` → `VendorBooth(booth_id)`
UNIQUE: (`event_id`, booth_id)

<u>Normalization:</u>

The resulting relations were analyzed to determine whether further normalization was necessary. The schema is already well-structured and does not require additional decomposition.

All relations satisfy **First Normal Form (1NF)** because each attribute stores atomic values only, and there are no repeating groups or multivalued attributes within a single relation.

The schema also satisfies **Second Normal Form (2NF)**. Relations with single-attribute primary keys automatically meet 2NF. For the relations with composite keys, such as `ParticipatesIn`, `SuppliesAt`, `RegistersFor`, `TeamAssignment`, and `VendorAssignment`, there are no partial dependencies. In each case, non-key attributes depend on the whole composite key rather than only part of it.

The schema further satisfies **Third Normal Form (3NF)** because there are no transitive dependencies among non-key attributes. Descriptive attributes in each relation depend directly on the primary key of that relation, and not on other non-key attributes.

The schema is also in **BCNF** under the stated key and uniqueness constraints. In each relation, the determinants of nontrivial functional dependencies are candidate keys or superkeys.

Therefore, no additional normalization is needed. The relations are already in a normalized form that supports data integrity, reduces redundancy, and preserves the structural constraints defined in the ER design.